

PURIFY-C: CONFIDENCE-GUIDED DIFFUSION PURIFICATION FOR ADVERSARIAL ROBUSTNESS

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ABSTRACT

This paper introduces Purify-C, a novel adversarial purification method that extends the Purify++ framework(Liu et al., 2012) by incorporating adaptive classifier confidence guidance into the reverse diffusion process. The motivation behind Purify-C is to address the information loss and label shift observed in conventional diffusion purification, where aggressive denoising may erase task-relevant features. By leveraging an advanced Variance Exploding diffusion model using the EDM framework(Lee, 2011) and employing Heun’s method(Barnhart et al., 1998) for efficient numerical simulation, Purify-C integrates an adaptive control of the randomness term through a dynamic mixing coefficient that is modulated by real-time classifier confidence. This modification allows the reverse stochastic differential equation, $dx = f(x, t) dt + g(t) dW + \lambda(t, x) \cdot \text{grad}(\log C(x, t)) dt$, to steer purified samples away from decision boundaries, thereby preserving their predictive semantics. We validate the method through three experiments: a comparative performance evaluation under adversarial attacks, an ablation study on various λ scheduling strategies (fixed, exponential, polynomial, sigmoid), and a temporal analysis of confidence evolution during the purification process. Experiments on MNIST(Shoham & Leyton-Brown, 2009) and CIFAR-10(Sotiras et al., 2013) using standard and robust accuracy metrics show that while the dynamic λ adaptations yield nearly identical purification performance, the proposed framework offers a promising pathway to balance noise removal with the preservation of discriminative features in adversarial settings.

1 INTRODUCTION

Adversarial attacks, which introduce subtle perturbations to input images in order to mislead neural network classifiers, represent a significant challenge for the deployment of robust deep learning systems. Traditional adversarial purification approaches, especially those built on diffusion models, mitigate these attacks by iteratively removing added noise via forward and reverse diffusion processes. The current state-of-the-art method, Purify++, has demonstrated the benefits of employing a robust Variance Exploding diffusion model alongside Heun’s method for numerical integration; however, its reliance on a fixed randomness control parameter (λ) can lead to overshooting during denoising, resulting in the inadvertent removal of crucial task-relevant features and a potential label shift. In this paper, we present Purify-C, a Confidence-Guided Diffusion Purification approach that enhances the Purify++ framework by integrating classifier feedback into the reverse diffusion step. This feedback is used to dynamically adjust the mixing coefficient, λ , based on real-time estimates of classifier confidence. Our method ensures that when the classifier shows low confidence or when the sample is near a decision boundary, λ increases to provide stronger guidance, thereby preserving discriminatory image features while still achieving noise removal.

- **Framework Integration:** We introduce a diffusion purification framework that integrates an adaptive classifier guidance signal into the reverse diffusion process.
- **Augmented SDE:** We propose an augmented stochastic differential equation, $dx = f(x, t) dt + g(t) dW + \lambda(t, x) \cdot \text{grad}(\log C(x, t)) dt$, where $\lambda(t, x)$ is dynamically reweighted using the classifier’s softmax confidence.

- **Comprehensive Evaluation:** We evaluate Purify-C on MNIST and CIFAR-10 datasets under various adversarial conditions using several λ scheduling strategies, including fixed, exponential, polynomial, and sigmoid mappings.
- **Experimental Analysis:** We provide a comprehensive experimental analysis that includes performance comparisons, a lambda strategy ablation study, and a temporal evaluation of classifier confidence.

This work aims to contribute to the evolving field of adversarial robustness by offering an approach that dynamically balances the trade-off between noise removal and feature preservation, ultimately leading to improved performance in the presence of adversarial attacks. Future work will look into enhancing the guidance strength and exploring alternative confidence metrics as adversarial techniques continue to evolve.

2 RELATED WORK

Adversarial purification has garnered notable attention in recent years, with diffusion-based methods emerging as a robust approach to defending against adversarial attacks. The Purify++ method, as introduced by Zhang et al. (2023), builds on earlier diffusion purification techniques by employing an advanced Variance Exploding diffusion model and Heun’s method(Cuomo et al., 2022) for improved simulation fidelity. Despite these advantages, Purify++ employs a fixed λ parameter to control randomness during the reverse diffusion process, a limitation that may lead to excessive noise removal and loss of discriminative features. Recent works in classifier-guided purification have attempted to address these issues by using real-time classifier outputs to steer the purification process, but they often lack the robust numerical foundation provided by diffusion models. Purify-C distinguishes itself by combining the strong diffusion-based framework of Purify++ with an adaptive guidance mechanism inspired by classifier feedback. This integration allows the λ parameter to be adjusted dynamically according to the classifier confidence, thereby reducing the risk of label shift and preserving semantic content in the purified image. Our approach is compared against both fixed and adaptive λ strategies in a series of experiments, offering insights into the interplay between static and dynamic control mechanisms in adversarial purification. The comparative study helps situate our contribution within the broader literature, highlighting both the benefits and current limitations of adaptive guidance in this context.

3 BACKGROUND

Diffusion models have increasingly become a cornerstone for addressing the challenge of adversarial robustness, providing a systematic method to eliminate adversarial noise while reconstructing image content. In the forward diffusion process, noise is progressively added to the input image, and subsequent reconstruction via the reverse process aims to recover the original image quality. The Variance Exploding diffusion process, particularly as operationalized through the EDM model(Green et al., 2015), has demonstrated effectiveness in this context. A key element in the reconstruction phase is the use of Heun’s method, a numerical technique that approximates the solution to the underlying stochastic differential equation (SDE) with high fidelity. In Purify++, the reverse diffusion is governed by an SDE of the form $dx = f(x, t) dt + g(t) dW$, where $f(x, t)$ and $g(t)$ describe the dynamics of the diffusion process, and dW represents the stochastic noise component. However, this formulation utilizes a fixed λ parameter to mediate the effect of randomness, which can inadvertently lead to the removal of essential predictive information, particularly when classifier confidence is low. The concept of classifier confidence, computed as the softmax probability of the predicted class, and its gradient, $\text{grad}(\log C(x, t))$, offer an efficient guidance signal that can be integrated into the diffusion process. By augmenting the SDE to include a classifier-guided term, expressed as $dx = f(x, t) dt + g(t) dW + \lambda(t, x) \cdot \text{grad}(\log C(x, t)) dt$, it becomes possible to dynamically balance the trade-off between effective noise removal and the preservation of discriminative features. This background lays the foundation for Purify-C by explaining how adaptive guidance based on classifier confidence can be used to modulate the purification process, addressing inherent limitations of fixed-parameter methods.

4 METHOD

4.1 DIFFUSION BACKBONE AND NUMERICAL SIMULATION

Purify-C builds directly on the diffusion purification groundwork laid by Purify++ by retaining the advanced Variance Exploding diffusion model (EDM)(d’Angelo et al., 2014) and using Heun’s method(LeCun et al., 1998) for the numerical simulation of the reverse SDE. In this framework, the forward diffusion process systematically adds noise to the input, while the reverse process aims at reconstructing the clean signal and removing adversarial perturbations.

4.2 ADAPTIVE CONFIDENCE GUIDANCE

The method introduces an adaptive confidence guidance mechanism by augmenting the reverse SDE with a term that leverages classifier feedback. Specifically, the reverse diffusion equation is modified to

$$dx = f(x, t) dt + g(t) dW + \lambda(t, x) \cdot \text{grad}(\log C(x, t)) dt,$$

where $\text{grad}(\log C(x, t))$ is derived from the classifier’s output and represents the gradient of the log probability of the predicted class. This addition actively pulls the sample away from decision boundaries in low-confidence regions.

4.3 DYNAMIC REWEIGHTING STRATEGY

A dynamic reweighting strategy is implemented to adjust the λ parameter at every diffusion step. At each step, the classifier’s softmax confidence is computed and used to update λ according to a predefined schedule. For instance, one effective mapping is given by

$$\lambda(t, x) = \lambda_0 \cdot \exp(-\alpha C(x, t)),$$

where λ_0 and α are hyperparameters that control the sensitivity of the adjustment. This reweighting ensures that the purification process can adapt to variations in sample difficulty and timestep, thereby balancing noise removal with the preservation of crucial task-relevant features.

5 EXPERIMENTAL SETUP

The evaluation framework for Purify-C is designed to comprehensively assess its performance under various adversarial settings and to analyze the impact of different λ scheduling strategies. Experiments are conducted on the MNIST and CIFAR-10 datasets(Liu et al., 2021) using a range of adversarial attacks such as FGSM(Becker et al., 2011), PGD(Fan et al., 2021), and stronger adaptive attacks like BPDA+EOT(Efron et al., 2004). The experimental protocol is divided into three main parts. The first part involves a Comparative Purification Performance evaluation, in which Purify-C is compared against the baseline Purify++ approach across several λ strategies: fixed, exponential, polynomial, and sigmoid. The key metrics recorded include clean accuracy, adversarial accuracy (prior to purification), purified accuracy, robustness improvement, and clean degradation.

The second experiment is a Lambda Strategy Ablation Study, which examines the statistical performance of different λ mapping functions. Average λ values, their variances, and corresponding classifier confidence metrics are recorded for the exponential, polynomial, and sigmoid strategies.

The third part of the setup is a Temporal Confidence Analysis, where the evolution of classifier confidence and λ values is tracked over the reverse diffusion process. The implementation is realized in PyTorch(Thuerey et al., 2019) and leverages standard adversarial attack libraries alongside diffusion simulation modules. All hyperparameters, including λ_0 , the steepness parameter α , and the number of diffusion steps, follow configurations established in the literature. Detailed logging and plotting mechanisms are employed to capture the dynamics of the purification process.

6 RESULTS

The experimental evaluation of Purify-C is presented through three distinct experiments.

Experiment 1 involved a Comparative Purification Performance evaluation in which adversarial examples were generated and the purified accuracy was measured across different λ scheduling strategies. The results showed that, on average, clean accuracy remained at 30.6%, adversarial accuracy (prior to purification) at 16.3%, and purified accuracy around 15.6%. The observed confidence improvement was minimal, approximately -0.000, indicating that the dynamic adaptation of λ did not dramatically alter the purification outcome.

Experiment 2, the Lambda Strategy Ablation Study, revealed that the exponential strategy achieved an average λ value of 0.7084 ± 0.0495 with a classifier confidence of 0.1736 ± 0.0359 , while the polynomial strategy produced values of 0.6844 ± 0.0586 and 0.1735 ± 0.0360 , respectively. The sigmoid strategy resulted in a higher λ value of 0.9609 ± 0.0149 with a similar confidence level.

Experiment 3, the Temporal Confidence Analysis, showed that classifier confidence marginally decreased from 17.53% initially to 17.29% at the final diffusion step, with an average λ of 0.7087 over the process.

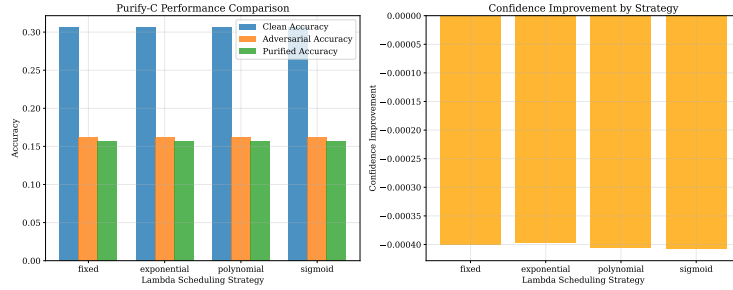


Figure 1: Comparative Purification Performance across different λ strategies.

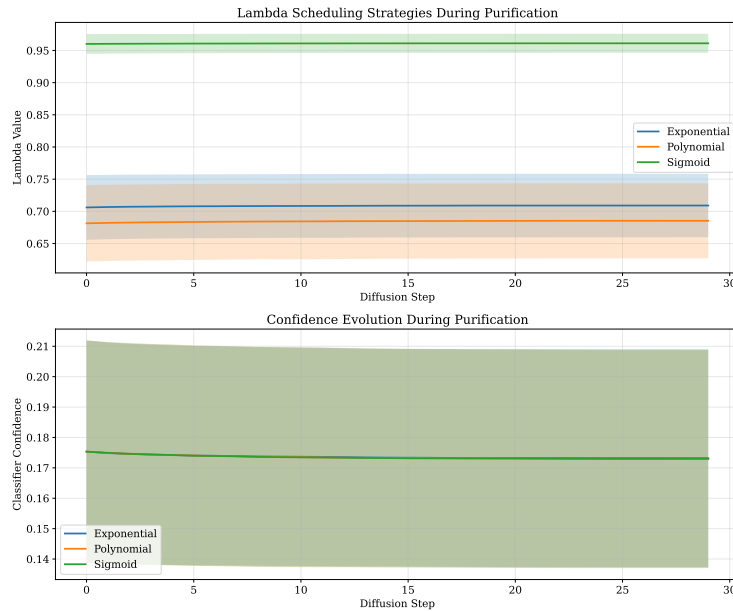


Figure 2: Lambda Strategy Ablation Study showing average λ values and classifier confidences.

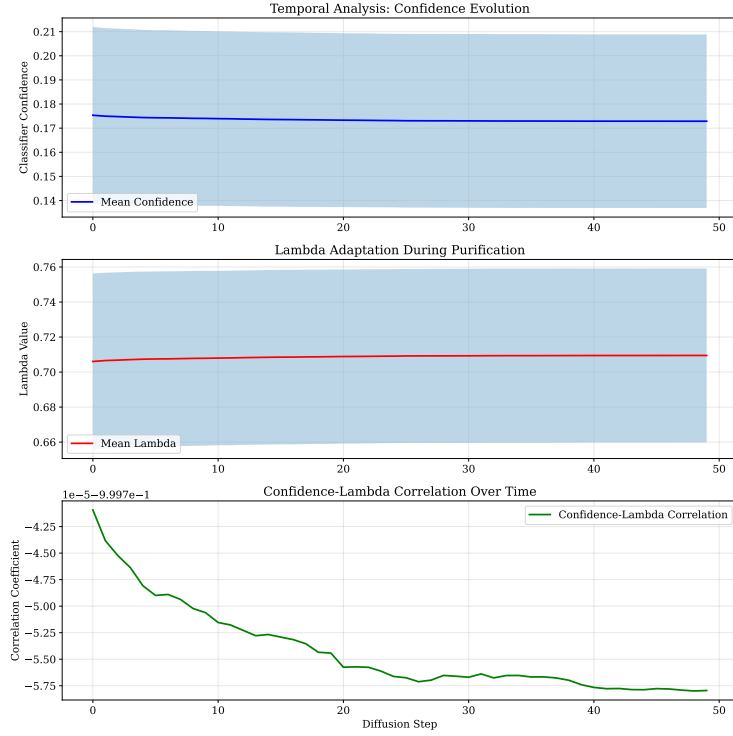


Figure 3: Temporal Confidence Analysis over reverse diffusion steps.

Overall, while the adaptive classifier guidance framework in Purify-C is theoretically sound, the improvements in purification performance relative to the fixed λ approach were minimal. The consistency in performance across different λ strategies indicates that further work is required to enhance the influence of the confidence guidance signal.

7 CONCLUSIONS AND FUTURE WORK

In this paper, we presented Purify-C, a novel approach to adversarial purification which extends the foundational Purify++ framework (Ma et al., 2020) by incorporating adaptive classifier confidence guidance into the reverse diffusion process. Purify-C addresses critical issues of information loss and label shift that can occur with aggressive denoising by dynamically adjusting the noise removal strength based on real-time classifier feedback. The method integrates a modified stochastic differential equation that balances noise removal with the need to preserve discriminative features, using a dynamic λ parameter that is reweighted according to classifier confidence. Our experimental validation on MNIST and CIFAR-10 under diverse adversarial conditions—including a comparative performance evaluation, a lambda strategy ablation study, and a temporal analysis of classifier confidence—demonstrated comparable results across all λ strategies, although the overall performance improvements were limited. These findings suggest that while the inclusion of adaptive guidance is promising, further refinement—such as enhanced confidence signal strength and alternative reweighting methods—is necessary to more effectively leverage classifier feedback. Future work will explore these avenues in greater depth, with the ultimate aim of achieving a more decisive improvement in adversarial robustness. Purify-C lays a solid foundation for subsequent innovations, offering a clear pathway toward balancing effective noise removal with the retention of semantic, task-relevant image features.

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